

Fermat's Last Theorem for Exponent Five

A Proof Method Using 5-Adic Valuations, the Golden Integer Ring, and Infinite Descent

An exposition based on the Lean 4 artifact `FLT5_Standalone_v4330_succ.lean`

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Abstract

This article explains, following the accompanying Lean 4 formalization artifact, a proof that the equation

$$x^5 + y^5 = z^5$$

has no solution in positive natural numbers. The central strategy is to reduce a hypothetical solution to a primitive one, isolate exactly the exceptional factors of 5 arising in a difference or a sum, and then pass to the quadratic integer ring $\mathbb{Z}[\varphi]$, where $\varphi^2 = \varphi + 1$. Because this ring is norm-Euclidean, relatively prime factors whose product is a fifth power are themselves fifth powers up to a unit. Unit classes modulo fifth powers reduce to five possibilities, four of which are eliminated by arithmetic modulo 5. The remaining zero unit class is ruled out by reconstructing arithmetic data of the same type from a fifth root while strictly decreasing a positive natural-valued measure. We describe the mathematical skeleton, the role of each invariant, and its correspondence with the principal Lean theorems.

Contents

1	The Main Theorem and the Place of the Proof	2
2	Primitive Reduction and Gap Factorization	2
2.1	Gap coordinates	3
3	Signed Branches and 5-Adic Valuations	3
3.1	Putting differences and sums into a common form	3
3.2	The residual contains exactly one factor of 5	4
4	Passage to the Golden Integer Ring	4
4.1	Reading the cyclotomic factor as a quadratic norm	4
4.2	Removing the ramified factor above 5 exactly once	5
5	Extraction of Fifth Powers and Unit Classes	5
5.1	Extracting a fifth power from relatively prime factors	5
5.2	Reducing the units to five classes	5
5.3	Eliminating the nonzero sectors	6
6	Arithmetic in the Zero Sector	6
7	Inversion Factorization and Infinite Descent	7

7.1	Diagonalizing the quartic form	7
7.2	A descent packet closed under re-entry	7
7.3	Strict decrease of the measure	8
8	Logical Closure of the Complete Proof	8
9	Proof-Engineering Lessons Visible in the Formalization	8
10	Conclusion	9
A	Guide to the Modules	9
B	Notes on Reading the Exposition	10

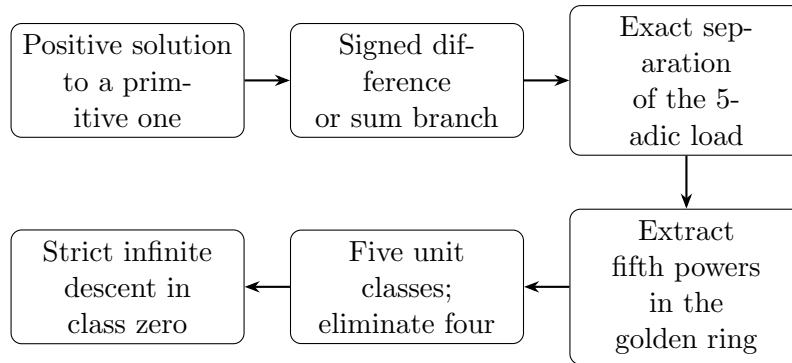
1 The Main Theorem and the Place of the Proof

Theorem 1.1 (Fermat’s Last Theorem for exponent five). *For all positive natural numbers x, y, z ,*

$$x^5 + y^5 \neq z^5.$$

In Lean, the equation is defined as `Fermat5Equation  $x\ y\ z$` , while the complete target is named `FLT5Target`. The development is closed by the final theorem `flt5Target` and the ordinary-argument wrapper `fermatFive_no_positive_solution`. Its scope is precisely positive natural numbers and exponent five; it does not directly assert the general FLT statement or a theorem about arbitrary integer solutions.

The proof can be summarized by the following flow.



Elementary factorization alone leaves an exceptional case in which the prime 5 occurs in both factors. A distinctive feature of this formalization is that it does not avoid the exception: congruences modulo 25, 5-adic valuations, binary quadratic forms of discriminant 5, and the ramification of 5 in $\mathbb{Z}[\varphi]$ are treated as one continuous proof route.

2 Primitive Reduction and Gap Factorization

Assume that a counterexample exists and remove common factors to obtain a primitive counterexample satisfying

$$\gcd(x, y) = 1, \quad x, y, z > 0, \quad x^5 + y^5 = z^5.$$

The Lean structure `CounterexamplePack` packages positivity, the relation $\gcd(x, y) = 1$, and the equation. The reduction from an arbitrary positive solution to such a structure is formalized by `exists_counterexamplePack_of_positive_fermat5`.

2.1 Gap coordinates

Set $g = z - y > 0$. Then

$$x^5 = z^5 - y^5 = g G_5(g, y),$$

where

$$G_5(g, y) = g^4 + 5g^3y + 10g^2y^2 + 10gy^3 + 5y^4. \quad (1)$$

This is the usual cyclotomic factor

$$z^4 + z^3y + z^2y^2 + zy^3 + y^4$$

after the substitution $z = g + y$.

The crucial congruences are

$$G_5(g, y) \equiv 5y^4 \pmod{g}, \quad (2)$$

$$G_5(g, y) \equiv g^4 \pmod{5}. \quad (3)$$

Primitivity also gives $\gcd(g, y) = 1$. Hence, by (2), every common prime divisor of g and $G_5(g, y)$ must be 5.

If $5 \nmid g$, then

$$\gcd(g, G_5(g, y)) = 1.$$

Since the product of these relatively prime numbers is a fifth power, one can split it as

$$g = a^5, \quad G_5(g, y) = b^5.$$

This elementary normal form corresponds to `branchB_fifth_power_factor_split`.

To finish the proof at this point, however, one would need a separate theorem saying that $G_5(g, y)$ is never a fifth power for every candidate. The formalization provides short contradictions from local primes through `CleanGN5Channel`, but the main route that uniformly closes all candidates is the signed 5-adic branch developed below.

3 Signed Branches and 5-Adic Valuations

3.1 Putting differences and sums into a common form

Modulo 5, one has $n^5 \equiv n \pmod{5}$. Combining this with a finite calculation modulo 25 shows that, after allowing the two left-hand variables to be exchanged, a primitive counterexample is routed into one of the following orientations:

- (i) Difference type: $5 \mid u$ and $5 \mid (w - v)$.
- (ii) Sum type: $5 \mid w$ and $5 \mid (u + v)$.

This is the content of `SignedBranchA0orientation` and `signedBranchA_normalForm_of_branchB`. In the sum case, the positive residual factor S_5 , defined in Lean as `SumGN5`, satisfies

$$u^5 + v^5 = (u + v)S_5(u, v).$$

Although the difference and sum representations are distinct, all invariants needed below have the same form.

3.2 The residual contains exactly one factor of 5

In either orientation, choose a carrier and a residual so that

$$\text{carrier} \cdot \text{residual} = \text{distinguished}^5.$$

The calculation modulo 25 gives

$$\text{residual} \equiv 5 \pmod{25}.$$

Consequently,

$$v_5(\text{residual}) = 1.$$

Because the complete product is a fifth power, additivity of the valuation gives

$$v_5(\text{carrier}) \equiv 4 \pmod{5}.$$

One further proves that the greatest common divisor of the two factors is exactly 5. Splitting the remaining relatively prime parts yields positive coprime natural numbers a, b such that

$$\text{carrier} = 5^4 a^5, \tag{4}$$

$$\text{residual} = 5b^5, \tag{5}$$

$$\text{distinguished} = 5ab, \tag{6}$$

$$\gcd(a, b) = 1, \quad 5 \nmid b. \tag{7}$$

These are the central data of **SignedFiveAdicPowerSplit**. The exponents 4, 1, 1 are not accidental: the sum of the valuations of the two factors must be a multiple of 5, and because the residual contributes exactly 1, the carrier contributes 4 mod 5.

4 Passage to the Golden Integer Ring

4.1 Reading the cyclotomic factor as a quadratic norm

Set

$$M = z^2 + y^2, \quad N = zy.$$

For the difference type, the residual becomes

$$G_5(z - y, y) = M^2 + MN - N^2.$$

For the sum type, the same formula holds after taking $N = -uv$. We therefore consider

$$R = \mathbb{Z}[\varphi], \quad \varphi^2 = \varphi + 1,$$

with conjugation and norm

$$\overline{m + n\varphi} = (m + n) - n\varphi, \quad N(m + n\varphi) = m^2 + mn - n^2.$$

The norm is multiplicative, and

$$4N(m + n\varphi) = (2m + n)^2 - 5n^2.$$

Thus discriminant 5 arises naturally.

Lean implements an element of the ring as an integer pair (m, n) , named **GoldenInt**, and defines addition, multiplication, conjugation, and the norm directly in coordinates. Simultaneous nearest-lattice rounding in rational coordinates constructs a division

$$x = qy + r, \quad |N(r)| < |N(y)|,$$

proving that R is norm-Euclidean. As a result, the proof may use greatest common divisors and the relevant consequences of unique factorization.

4.2 Removing the ramified factor above 5 exactly once

Let

$$\tau = 2 + \varphi, \quad N(\tau) = 5.$$

The golden integer $\alpha = M + N\varphi$ obtained from (4.1) and (5) satisfies $N(\alpha) = 5b^5$. Divisibility of its diagonal coordinate gives a factorization

$$\alpha = \tau\beta.$$

After removing τ , the element β satisfies

$$N(\beta) = b^5, \tag{8}$$

$$\beta_2 = -5^7 a^{10}, \tag{9}$$

$$5 \nmid N(\beta), \tag{10}$$

and it is not divisible by τ again. Here β_2 denotes the second coordinate in the basis $1, \varphi$. Concentrating the large 5-adic load in this coordinate makes the later elimination of unit classes possible.

Moreover, every common divisor of β and $\bar{\beta}$ is a unit. Indeed, the norm of a common divisor divides both $N(\beta) = b^5$ and $N(\beta - \bar{\beta}) = -5\beta_2^2$. By (7) and $5 \nmid b$, only absolute norm 1 remains possible.

5 Extraction of Fifth Powers and Unit Classes

5.1 Extracting a fifth power from relatively prime factors

We have

$$\beta\bar{\beta} = b^5,$$

and the two factors are relatively prime. Unique factorization in the Euclidean domain therefore gives a unit $\varepsilon \in R^\times$ and an element $\gamma \in R$ such that

$$\beta = \varepsilon\gamma^5.$$

The general result is `goldenCoprimeFactorOfFifthPower`; its application to the FLT5 data is `signedGoldenFifthPowerUpToUnitCore`.

5.2 Reducing the units to five classes

The units of the golden integer ring can be classified by an elementary descent that reduces a coordinate-size measure. The exact form needed downstream is

$$\varepsilon = \varphi^i \delta^5, \quad i \in \{0, 1, 2, 3, 4\}.$$

Because the exponent 5 is odd, the sign can be absorbed into δ^5 . Thus (5.1) reduces to five algebraic sectors:

$$\beta = \varphi^i \theta^5, \quad i = 0, 1, 2, 3, 4.$$

Here “sector” means a residue class of the unit group modulo fifth powers, not a geometric angular region.

5.3 Eliminating the nonzero sectors

Write $\theta = r + s\varphi$, and define

$$F(r, s) = r^5 + 10r^3s^2 + 10r^2s^3 + 10rs^4 + 3s^5, \quad (11)$$

$$H(r, s) = r^4 + 2r^3s + 4r^2s^2 + 3rs^3 + s^4. \quad (12)$$

Then

$$\theta^5 = F(r, s) + 5sH(r, s)\varphi.$$

Furthermore,

$$F(r, s) \equiv r + 3s \pmod{5}, \quad N(\theta) \equiv (r + 3s)^2 \pmod{5}.$$

The second coordinate of $\varphi^i \theta^5$ takes the following form.

i	Second coordinate
0	$5sH$
1	$F + 5sH$
2	$F + 2(5sH)$
3	$2F + 3(5sH)$
4	$3F + 5(5sH)$

Equation (9) implies $5 \mid \beta_2$. For $i = 1, 2, 3, 4$, the table therefore gives $5 \mid F(r, s)$, and (5.3) yields $5 \mid N(\theta)$. Applying the norm to (5.1), however, gives $N(\theta) = \pm b$, contradicting $5 \nmid b$. All four nonzero sectors are therefore impossible.

Only the zero sector remains:

$$\boxed{\beta = \gamma^5}.$$

6 Arithmetic in the Zero Sector

Write $\gamma = r + s\varphi$. Comparing second coordinates gives

$$sH(r, s) = -5^6 a^{10}.$$

We also have

$$N(\gamma) = \pm b, \quad 5 \nmid b.$$

Primitivity yields $\gcd(|r|, |s|) = 1$, and consequently

$$\gcd(|s|, |H(r, s)|) = 1.$$

The congruence

$$H(r, s) \equiv N(r + s\varphi)^2 \pmod{5}$$

also gives $5 \nmid H(r, s)$. Hence all factors of 5 in (6) lie on the s -side. Since the remaining relatively prime product is a tenth power, there exist $c, d \in \mathbb{N}$ such that

$$|s| = 5^6 c^{10}, \quad (13)$$

$$|H(r, s)| = d^{10}. \quad (14)$$

Moreover, $H(r, s) \geq 0$, while the right-hand side of (6) is negative, so

$$s = -5^6 c^{10}, \quad H(r, s) = d^{10}.$$

At this point the original three-variable FLT equation has been transformed into a special primitive equation for a binary quartic form. The structure `GoldenZeroSectorCandidate` retains all positivity, coprimality, norm, sign, and tenth-power-splitting information needed for the transformation.

7 Inversion Factorization and Infinite Descent

7.1 Diagonalizing the quartic form

Set

$$X = 2r + s.$$

A direct calculation gives

$$16H(r, s) = X^4 + 10X^2s^2 + 5s^4.$$

Now define

$$U = X^2 + 5s^2, \quad W = 4d^5, \quad A = U - W, \quad B = U + W.$$

Combining (13), (14), and (7.1) yields

$$AB = 4Q^5, \quad Q = 5^5c^8, \tag{15}$$

$$B - A = 8d^5. \tag{16}$$

Moreover, $0 < A < B$. The factors are then split exactly according to their 2-adic valuations. The formalization distinguishes one odd branch and two even branches and uses an auxiliary congruence modulo 11 in the odd branch. These branches are not the final objective; they are arithmetic certificates that permit a safe re-entry into the following recursive invariant.

7.2 A descent packet closed under re-entry

Write the essential descent data as $\eta = R + S\varphi$. Schematically, it satisfies

$$S = \pm 5t^5, \tag{17}$$

$$H(R, S) = D^5, \tag{18}$$

$$\gcd(|R|, |S|) = 1, \quad 5 \nmid N(\eta). \tag{19}$$

This is the structure `GoldenZeroSectorDescentPacket`.

Consider the “golden lift”

$$L(\eta) = (R^2 + RS + S^2) + S^2\varphi.$$

A coordinate calculation gives

$$N(L(\eta)) = H(R, S),$$

and $L(\eta)$ is relatively prime to its conjugate. Applying (18) and fifth-power extraction in the Euclidean domain once more, the nonzero unit classes are eliminated modulo 5, giving an element $\theta = p + q\varphi$ such that

$$L(\eta) = \theta^5.$$

Comparing second coordinates gives

$$S^2 = 5qH(p, q).$$

Splitting the two factors on the right again by coprimality produces

$$q = 5u^5, \quad H(p, q) = v^5.$$

Thus θ itself supplies a new descent packet of the same type as (17)–(19).

7.3 Strict decrease of the measure

Take the absolute value of the second coordinate as the measure:

$$\mu(\eta) = |S| \in \mathbb{N}_{>0}.$$

Using (7.2) together with the nonnegative identity (7.1), one obtains

$$|q| < |S|,$$

and therefore

$$\mu(\theta) < \mu(\eta).$$

Every descent packet consequently produces another packet with the same properties and a strictly smaller positive natural-valued measure. Such an infinite descending sequence cannot exist in the natural numbers, giving the contradiction.

In Lean, `GoldenZeroSectorDescentPacket.strictDescent` supplies one descent step, and `goldenZeroSectorDescentPacket_false` closes the argument by strong induction on the natural-number measure. The construction assumes no final FLT theorem; it uses only the preceding local lemmas, so the reasoning is non-circular.

8 Logical Closure of the Complete Proof

Infinite descent rules out every zero-sector candidate. Hence all five unit classes are impossible, eliminating the possibility that β is a unit times a fifth power. This rules out both signed difference and sum types and therefore every primitive counterexample. Finally, reversing the reduction from an arbitrary positive solution to a primitive counterexample proves the main theorem.

The principal closure declarations in the formalization are as follows.

Lean declaration	Mathematical role
<code>goldenZeroSectorCandidate_false</code>	Rules out a zero-sector candidate by strict infinite descent.
<code>goldenZeroSectorArithmeticExclusion</code>	Proves the public zero-sector arithmetic interface unconditionally.
<code>goldenUnitClassesModFifth</code>	Classifies golden units into five classes modulo fifth powers.
<code>counterexamplePackRefuter_of_unitClasses_of_zeroArithmetic</code>	Rules out primitive counterexamples from the unit classification and zero-sector exclusion.
<code>exists_counterexamplePack_of_positive_fermat5</code>	Reduces an arbitrary positive solution to a primitive counterexample.
<code>flt5Target</code>	States the impossibility of $x^5 + y^5 = z^5$ in positive natural numbers.

9 Proof-Engineering Lessons Visible in the Formalization

Localize the exceptional prime 5. By (2), the only prime that can spoil the ordinary coprime factorization is 5. Working modulo 25 strengthens the residual valuation from “at least one” to “exactly one.”

Manage natural-number subtraction early. For the gap $z - y$, the inequality $y < z$ is proved first; for the sum orientation, the always-positive `SumGN5` is used. Sign issues are therefore handled safely over the natural numbers until the proof deliberately passes to an integer ring.

Preserve information in packets. Each structure records not only existence but also positivity, provenance, congruences, coprimality, valuations, and coordinate identities. Later stages do not need to reopen earlier proofs, making dependencies and non-circularity explicit.

Construct unique factorization from a coordinate model. Rather than treating $\mathbb{Z}[\varphi]$ as a black-box number field, the development implements it as integer pairs and proves norm-Euclidean division. It then uses general gcd-monoid results to extract fifth powers, separating FLT-specific arithmetic from general commutative algebra.

Express infinite descent as well-founded induction. The classical statement that a smaller solution is obtained is made precise by the natural-valued measure $|S|$ and a strict inequality. Infinite descent then becomes strong induction on the natural numbers in Lean.

10 Conclusion

The difficult point in the exponent-five case is not merely the factorization

$$(z - y)G_5(z - y, y) = x^5,$$

but the complete control of how the exceptional prime 5 is distributed between the factors. Arithmetic modulo 25 fixes the 5-adic valuation of the residual at 1. The norm in the golden integer ring converts the cyclotomic factor into a quadratic product. Euclideanity extracts fifth powers from that product, and unit classification reduces infinitely many possibilities to five. Four disappear immediately modulo 5; the last regenerates arithmetic data of the same type with a smaller measure and therefore disappears by infinite descent.

The proof thus consists of five layers:

$$\begin{aligned} &\text{elementary factorization} \longrightarrow \text{5-adic valuation} \longrightarrow \text{quadratic integer ring} \\ &\longrightarrow \text{unit classes} \longrightarrow \text{infinite descent.} \end{aligned}$$

The accompanying Lean artifact decomposes each layer into an independent interface with checkable invariants and ultimately closes the exponent-five statement for positive natural numbers in a form that can be checked by the kernel.

A Guide to the Modules

Generated source modules	Contents
<code>Basic</code> , <code>GN5</code> , <code>Reduction</code>	Primitive counterexamples, gaps, the cyclotomic factor, and coprimality away from 5.
<code>SignedBranchA</code> , <code>SignedFiveAdic</code> , <code>SignedFiveAdicPowerSplit</code>	Signed difference and sum routes, arithmetic modulo 25, exact 5-adic valuations, and equations (4)–(6).
<code>SquareGoldenBridge</code> , <code>SignedSquareGoldenExceptional</code> , <code>GoldenOrder</code> , <code>GoldenDivisibility</code> , <code>GoldenEuclidean</code>	The bridge from the cyclotomic factor to a quadratic norm of discriminant 5. Coordinate implementation of the golden integer ring, norm, conjugation, units, and Euclidean division.
<code>SignedGoldenRamifierStripped</code> , <code>SignedGoldenConjugateCoprime</code>	Removal of the norm-five ramifier τ and relative primality with the conjugate factor.

Generated source modules	Contents
GoldenCoprimeFactor, GoldenUnitClassification SignedGoldenSectorArithmetic, SignedGoldenZeroSector SignedGoldenZeroSectorInversion, SignedGoldenZeroSectorFactorization SignedGoldenZeroSectorDescent, SignedGoldenZeroSectorFinal Main	Fifth-power extraction from relatively prime factors and classification into five unit classes. Elimination of the four nonzero unit classes and tenth-power splitting in the zero class. Inversion of the quartic form, positive factors A, B , and certificates for the 2-adic branches. The recursive invariant, strict measure decrease, and closure by strong induction. Assembly of the conditional interfaces and presentation of <code>flt5Target</code> .

B Notes on Reading the Exposition

This article reconstructs the proof architecture of the specified generated Lean file for a human reader. For reasons of space, it omits the complete finite residue calculations modulo 25 and 11, every case of the 2-adic branching, the rounding inequalities in the Euclidean algorithm, and the detailed type conversions between packets. These appear as individual lemmas in the formalization artifact. Moreover, saying that Lean checks the proof means that the source compiles in the corresponding Lean/Mathlib environment and that the theorem terms are type-checked by the kernel. It does not by itself assert historical novelty, external peer review, or the general-exponent form of FLT.

References

- [1] `FLT5_Standalone_v4330_succ.lean`, generated standalone Lean 4 artifact, DkMath/FLT/Five modules, source commit recorded in the file header.
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